TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Industrial LLM-based Assistant for Intelligent Machine Data Analysis, Predictive Reporting, and Process Optimization (Factory-GPT)			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication			
	Medium Category	Small Category	Medium Category	Small Category
	Software	Software Solutions		
Concept of Proposed Technology	Development of a Context-Aware Industrial Large Language Model (Ind-LLM) designed to interface directly with manufacturing equipment (PLC/SCADA/MES). Unlike generic models, this "Factory GPT" will: 1. Ingest and interpret raw machine data (time-series, logs, error codes) via protocols like OPC UA. 2. Contextualize data using RAG (Retrieval-Augmented Generation) [1] with technical manuals and historical maintenance records. 3. Generate human-readable reports and actionable optimization recommendations for operators in natural language.			
Rationale for Joint R&D Project	Korean SMEs possess advanced hardware and manufacturing data but may lack the specialized AI data science workforce to analyze it deeply. The Fraunhofer IPT provides deep process knowhow and is engaged in AI technologies for industrial implementation [2], while the Korean SME can provide the specific domain knowledge, real-world production data, and the testbed environment necessary to validate the model's accuracy and safety in a factory setting [3].			

Key Words	1. Large Language Models (LLM) 2. Predictive Maintenance 3. Natural Language Interface 4. Automated Reporting 5. Process Optimization

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level 2	**End of the Project**	Level 5
R&D Objectives	1. Develop a multimodal LLM architecture capable of processing both text (e.g. manuals) and structured data (e.g. sensor values). 2. Implement an intelligent query interface where operators can ask "Why did the machine stop?" and receive a root-cause analysis. 3. Ensure data privacy and security by developing on-premise or hybrid-cloud deployment strategies suitable for sensitive industrial data.			
R&D Target Specifications	1. Analysis Accuracy: > 90% correct identification of error causes based on log data. 2. Response Latency: < 5 seconds for complex query generation. 3. Hallucination Rate: < 2% (strict constraints for industrial safety [4]). 4. Integration: Support for at least 3 major industrial protocols (e.g., OPC UA, MQTT, Modbus).			
R&D Plan	Year 1 • Data collection from SME partner • Pre-processing and cleaning of industrial datasets • Selection and fine-tuning of base model (e.g., Llama 3 or similar) on industrial corpus. Year 2 • Development of the RAG (Retrieval-Augmented Generation) pipeline • Integration of the "Factory GPT" into a usable dashboard for the SME. • Pilot testing and iterative reinforcement learning from human feedback (RLHF) by factory operators.			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Design the AI model architecture and RAG framework. • Perform fine-tuning of Large Language Models on industrial datasets. • Develop the algorithm for translating numerical sensor data into semantic text embeddings. • Ensure scientific validation of the model's reasoning capabilities.
Expected R&R of Korean SME Partner	• Provide historical machine data, error logs, and maintenance manuals. • Define specific use-cases and "pain points" in current daily operations. • Integrate the prototype into their factory. • Conduct field testing and evaluate the utility of the generated reports.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• Creation of a specialized "Industrial LLM" that outperforms generic models (like standard ChatGPT) in technical troubleshooting accuracy. • Development of a SaaS or On-Premise software solution that can be sold to other manufacturers facing skilled labor shortages. • Faster error diagnosis leads to higher OEE (Overall Equipment Effectiveness). • The system preserves the knowledge of senior engineers and makes it accessible to junior staff via the chat interface. • Automated reporting saves engineers time, previously spent on manual data analysis.

6. References

[1] Lewis, P., et al. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.* (Foundational framework for RAG). https://doi.org/10.48550/arXiv.2005.11401

[2] Leutner, L. et al. (2025). *Generative AI for production and business operations.* Fraunhofer IPT. *https://doi.org/10.24406/publica-3781*

[3] Li, Y., et al. (2024). *Large Language Models for Manufacturing.* *https://doi.org/10.48550/arXiv.2410.21418*

[4] ISO/IEC 23894:2023. *Information technology — Artificial intelligence — Guidance on risk management.* https://www.iso.org/standard/77304.html

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization